

ble. Data delivered safely can be harvested by Trojans at the exposed endpoints. Whether a hacker's goal is vandalism or illicit profit, the costs can be enormous. Computer Economics, an independent research firm, estimated global financial damage from malicious code in 2000 at \$17.1 billion. mi2g, a London-based e-commerce research and development company, put the mark even higher, at \$20 billion.

New technologies have triggered a paradigm shift in network security. In the old network model, almost all PCs connected to the Internet via a central gateway. Guarding the gateway effectively created a defensive perimeter. This model is no longer adequate.

First, while corporate networks shored up their security with centralised firewalls, anti-virus and intrusion detection, hackers exposed other vulnerabilities. Second, the explosion of remote and mobile users with always-on, broadband Internet connections to the network means most networks now have hundreds, or even thousands, of vulnerable 'backdoors'.

The Gartner Group noted, "Broadband connections are rife with threats to remote devices. Viruses, Trojan horses, zombies, keystroke monitoring, file shares and denial-of-service attacks all threaten the remote machine and, by extension, put the enterprise's IT resources at risk." Microsoft and others were hacked in this way. Incursions of this sort can quickly turn into high-profile PR disasters, or worse, go undetected for months before being exposed. Vulnerability has expanded exponentially. As companies go from single gateways to thousands of Internet connected endpoints, the number of vulnerabilities for networks has exploded. IDC reported the number of remote users in the year 2000 at 39 million and growing nine percent annually.

This accelerating trend is creating even more back doors. In addition, laptop users physically bypass the firewall every day, and wireless networks have no definable boundaries. Effectively, the network perimeter has disappeared. Hackers have taken notice, and so have government regulators: recent legislation requiring tighter security